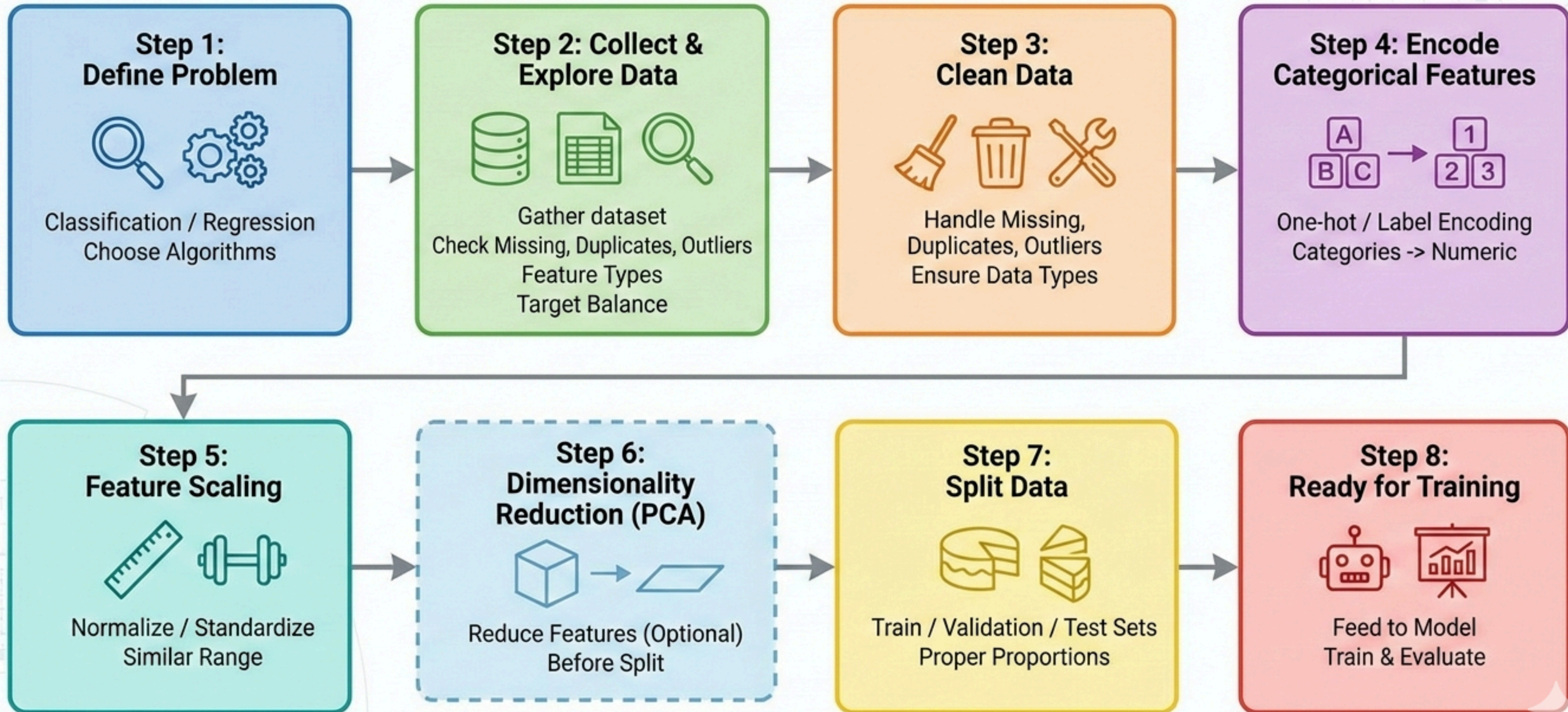


MACHINE LEARNING CONCEPTUAL FLOW



Step 1: Define the Problem

- Decide what you want to predict: classification (categories) or regression (continuous numbers).
- Think about which ML algorithms might suit your problem.

Step 2: Collect & Explore Data

- Gather your dataset (CSV, Excel, database, API).
- Examine the data:
 - Missing values
 - Duplicates
 - Outliers
 - Feature types (numeric, categorical)
 - Balance of target classes (for classification)

Step 3: Clean the Data

- Handle missing values: remove or impute with mean/median/mode.
- Remove duplicates.
- Handle outliers (remove or cap extreme values).
- Ensure proper data types for each column.

Step 4: Encode Categorical Features

- Convert categories into numbers:
 - One-hot encoding for nominal categories (no order)
 - Label/ordinal encoding for ordered categories

Step 5: Feature Scaling

- Scale numeric features so all are on a similar range.
- Needed for algorithms sensitive to magnitude:
 - Gradient-based: Linear/Logistic Regression, SVM
 - Distance-based: KNN, K-Means
- Not needed for tree-based algorithms (Random Forest, Decision Tree)
- Methods:
 - Normalization → values between 0–1
 - Standardization → zero mean, unit variance

Step 6: Dimensionality Reduction (PCA)

- Use PCA before splitting to reduce features if the dataset has many features.
- Steps:
 - a. Scale features (required for PCA).
 - b. Apply PCA to reduce dimensions while keeping most variance.
- PCA reduces overfitting, speeds up training, and can simplify visualization.

Important: If you are very strict about preventing data leakage, fit PCA only on the training set and apply the same transformation to validation/test sets after the split.

Step 7: Split Data

- Divide the processed dataset into:
 - Training set → to train the model
 - Validation set → to tune hyperparameters
 - Test set → to evaluate final performance
- Ensure proper proportions (e.g., 70% train, 15% validation, 15% test).

Step 8: Ready for Training

- After cleaning, encoding, scaling, (optional PCA), and splitting, the data is now ready.
- Feed the data into your chosen ML model for training and evaluation.

- Define problem → classification/regression
- Collect & explore → check missing, duplicates, outliers, feature types
- Clean → handle missing, duplicates, outliers
- Encode → categorical → numeric
- Scale → normalization or standardization
- Dimensionality reduction → PCA (optional, before split)
- Split → train / validation / test
- Feed to model → train & evaluate